FarmShift Agronomic Advisory & Regional Intelligence

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1. Farm Profile Baseline

Farm Name	Report Test Holding	**Region**	Nagpur
Total Area	5.0 Hectares	**Land Type**	Irrigated
Kharif Crop	Cotton	**Rabi Crop**	Wheat
Water Source	Canal (High)	**Soil Texture**	Deep Black Vertisol [Tested]

Operational Constraints: Drainage required in July

2. Evidence-Based Agronomic Insights

Recommendations synthesized using ICAR-CICR Nagpur, PDKV Akola, and IMD benchmarks. Advisories are grounded strictly on verified regional soil moisture physics and crop rotation dynamics.

Category	Priority	Finding & Advisory	Institutional Citation
Soil & Drainage	HIGH	Adopt Broad Bed and Furrow (BBF) configuration to relieve root asphyxiation during heavy July rains and retain subsoil moisture into October.	ICAR-CICR Nagpur Vertisol Drainage Advisory (2024)
Water Budget	HIGH	Mid-season dry spells in Vidarbha (mean 14-18 days) require in-situ straw mulching (3 t/ha) at 30 DAS to extend moisture buffer.	PDKV Akola Dryland Agriculture Centre Bulletin
Crop Protection	MEDIUM	Terminate continuous cotton residues by Dec 31 to interrupt Pink Bollworm pupae diapause; rotate with non-host legume.	All India Coordinated Cotton Improvement Project

3. Crop Rotation Analysis

System	Benefits	Trade-offs	Est. Return (INR/ha)
Cotton -> Chickpea -> Sunnhemp	Breaks pest cycle; fixes 30 kg N/ha; Sunnhemp green manure revitalizes clay structure.	Requires 1-2 light irrigations in Dec; pulse price fluctuation.	**Rs 52,000 / ha**
Soybean -> Wheat -> Summer Mung	300% cropping intensity; high seasonal liquidity; timely wheat sowing.	High pump water requirement; summer heat risk during May bloom.	**Rs 68,000 / ha**
Pigeonpea + Soybean (1:4) Intercrop	Deep taproots penetrate subsoil hardpan; drought-resilient hedge.	Long 180-day cycle prevents dense winter crop.	**Rs 46,000 / ha**

4. Scientific Limitations & Missing Inputs

• **Weather:** Deterministic long-range daily forecasts are not implied; analyses utilize 30-year IMD gridded statistics.
• **Soil Testing:** Laboratory chemical testing card status is noted. If unverified, nutrient recommendations are regional estimates.
• **Groundwater:** On-farm water metering data is recommended to refine irrigation scheduling during summer.